Comparing Bootstrap by Location and by Event

Comparison of Location-Stratified and Complete-Event Bootstrap Control Limits

A sensitivity analysis was conducted using all available Equipment B data to evaluate whether preserving the observed within-event relationships among sampling locations materially affects the estimated control limits. The comparison was performed separately for DAR, CAR, and Mic.

For each residue type, ND observations were replaced by the LOD, or by the LOQ when the LOD was unavailable. Each usable observation was then normalized by its corresponding USL $Z_{tr} = 100X_{tr}/\text{USL}_{tr}$, where Z_{tr} denotes the normalized result from sampling location r in cleaning event t.

Location-stratified bootstrap

For every sampling location r, define the empirical location stratum $G_r = \{Z_{1r}, \dots, Z_{n_r r}\}$.

For bootstrap replication $b = 1, \dots, B$:

1. Sample one observation Z^*_{br} with replacement from each location stratum G_r.
2. Combine the independently sampled location-specific observations to form one synthetic cleaning event.
3. Calculate the synthetic event-level location statistic $L_b^{(\text{location})} = \text{median}_r\, Z^*_{br}$.
4. Calculate the synthetic event-level scale statistic $S_b^{(\text{location})} = \text{median}_r\, |Z^*_{br} - L_b^{(\text{location})}|$.

No smoothing noise is added in this sensitivity analysis. Therefore, the location-stratified bootstrap differs from the complete-event bootstrap only in whether the historical within-event relationships among sampling locations are preserved.

The location-stratified bootstrap control limits are estimated as $UCL_L^{(\text{location})} = Q_{1-\alpha}\left\{L_1^{(\text{location})}, \dots, L_B^{(\text{location})}\right\}$, and $UCL_S^{(\text{location})} = Q_{1-\alpha}\left\{S_1^{(\text{location})}, \dots, S_B^{(\text{location})}\right\}$, where $\alpha = 1/\text{ARL}_0$.

Complete-event bootstrap

For each historical cleaning event t, calculate $L_t = \text{median}_r\, Z_{tr}$ and $S_t = \text{median}_r\, |Z_{tr} - L_t|$.

For bootstrap replication $b = 1, \ldots, B$:

1. Sample one complete historical cleaning event with replacement from the available Equipment B events.
2. Retain all sampling-location observations within the selected event as a single unit.
3. Record the corresponding event-level statistics $L_b^{(\text{event})}$ and $S_b^{(\text{event})}$.

This resampling procedure preserves the observed relationships among sampling locations within each historical cleaning event. The complete-event bootstrap control limits are estimated as $UCL_L^{(\text{event})} = Q_{1-\alpha}\left\{L_1^{(\text{event})}, \ldots, L_B^{(\text{event})}\right\}$, and $UCL_S^{(\text{event})} = Q_{1-\alpha}\left\{S_1^{(\text{event})}, \ldots, S_B^{(\text{event})}\right\}$.

Comparison of bootstrap control limits

The absolute differences between the two bootstrap procedures are defined as $\Delta_L = UCL_L^{(\text{location})} - UCL_L^{(\text{event})}$ and $\Delta_S = UCL_S^{(\text{location})} - UCL_S^{(\text{event})}$.

The corresponding percentage differences, using the complete-event bootstrap limits as the reference, are $\Delta_L(\%) = 100\frac{UCL_L^{(\text{location})} - UCL_L^{(\text{event})}}{UCL_L^{(\text{event})}}$ and $\Delta_S(\%) = 100\frac{UCL_S^{(\text{location})} - UCL_S^{(\text{event})}}{UCL_S^{(\text{event})}}$.

When $UCL_S^{(\text{event})} = 0$, the percentage difference for the scale limit is undefined and only the absolute difference is reported. Because the bootstrap replications generated by the two procedures have no natural one-to-one correspondence, individual simulated L_b and S_b values are not paired or tested. The sensitivity assessment focuses on the resulting UCL_L and UCL_S values.

	ResidueType	n_events	n_locations	n_distinct_event_location_sets	B
1	DAR	400	5	1	2000000
2	CAR	91	3	1	2000000
3	Mic	58	3	1	2000000

	target_ARL0	alpha	UCL_L_location	UCL_L_event	Delta_UCL_L
1	740	0.001351351	19.42683	19.42683	0.000000
2	740	0.001351351	24.78177	31.02813	-6.246363
3	740	0.001351351	4.00000	4.00000	0.000000

	Percent_Delta_UCL_L	UCL_S_location	UCL_S_event	Delta_UCL_S	Percent_Delta_UCL_S
1	0.000	6.136133	19.292683	-13.156550	-68.195
2	-20.131	14.859360	5.247333	9.612027	183.179
3	0.000	4.000000	0.000000	4.000000	NA

Table 1: Comparison of location-stratified and complete-event bootstrap upper control limits for Equipment B.

	UCL_L				UCL_S			
Residue	Loc.	Event	Δ	Δ%	Loc.	Event	Δ	Δ%
DAR	19.4268	19.4268	0.0000	0.000	6.1361	19.2927	-13.1566	-68.195
CAR	24.7818	31.0281	-6.2464	-20.131	14.8594	5.2473	9.6120	183.179
Mic	4.0000	4.0000	0.0000	0.000	4.0000	0.0000	4.0000	NA

Simulation Procedure for Comparing Complete-Event Simulation and Location-Stratified Bootstrap

A simulation study was conducted to examine whether location-stratified bootstrap resampling can approximate the control limits obtained from complete events when sampling locations are independent.

Data-generating model

Each cleaning event was assumed to contain measurements from 10 sampling locations. The location measurements were generated independently from normal distributions with location-specific means and a common standard deviation: $X_{tr} \sim N(\mu_r, 2^2)$, where X_{tr} represents the observation from location r in event t, and $\mu_r = 1 + 0.1r, \qquad r = 1, \dots, 10$. Therefore, the 10 location-specific distributions were $X_{t1} \sim N(1.1, 2^2)$, $X_{t2} \sim N(1.2, 2^2)$, ... $X_{t,10} \sim N(2.0, 2^2)$.

Measurements from different sampling locations and different events were generated independently.

For any simulated event t, the event-level location and scale statistics were defined as $L_t = \text{median}_{r=1,\dots,10} X_{tr}$ and $S_t = \text{median}_{r=1,\dots,10} |X_{tr} - L_t|$. Here, S_t is the unscaled median absolute deviation within event t.

Complete-event simulation

First, 20,000 complete events were generated directly from the specified location distributions. For each simulated event, one observation was independently generated from each of the 10 location-specific normal distributions: $\mathbf{X}_t = (X_{t1}, \dots, X_{t,10})$.

The corresponding event-level statistics L_t and S_t were then calculated for each of the 20,000 simulated events.

Let $\alpha = 1/\text{ARL}_0$, where $\text{ARL}_0 = 740$. The complete-event limits were estimated as $UCL_L^{(\text{event})} = Q_{1-\alpha}(L_1, \dots, L_{20000})$ and $UCL_S^{(\text{event})} = Q_{1-\alpha}(S_1, \dots, S_{20000})$, where $1 - \alpha = 1 - \frac{1}{740} \approx 0.9986486$.

These limits approximate the control limits under the known event-generating model.

Construction of historical location strata

Separately, a historical reference dataset containing 100 events was generated from the same model. Each historical event contained one observation from each of the 10 sampling locations.

For each sampling location r, the 100 historical observations were used to construct an empirical location stratum: $G_r = \{X_{1r}, X_{2r}, \dots, X_{100,r}\}$, for $r = 1, \dots, 10$. Thus, G_r represents the empirical distribution of observations from sampling location r based on the 100 historical events.

Location-stratified bootstrap

A total of $B = 20{,}000$ synthetic events were generated using location-stratified bootstrap resampling.

For each bootstrap replication $b = 1, \dots, B$:

1. One observation X^*_{br} was sampled with replacement from each empirical location stratum G_r.
2. The 10 independently sampled location values were combined to form one synthetic event: $\mathbf{X}^*_b = (X^*_{b1}, \dots, X^*_{b,10})$.
3. The event-level location statistic was calculated as $L_b^{(\text{location})} = \text{median}_{r=1,\dots,10}\, X^*_{br}$.
4. The event-level scale statistic was calculated as $S_b^{(\text{location})} = \text{median}_{r=1,\dots,10}\, |X^*_{br} - L_b^{(\text{location})}|$.

No smoothing noise was added. Therefore, the location-stratified bootstrap sampled only from the empirical location-specific distributions obtained from the 100 historical events.

The location-stratified bootstrap limits were estimated as $UCL_L^{(\text{location})} = Q_{1-\alpha}\left(L_1^{(\text{location})}, \dots, L_B^{(\text{location})}\right)$ and $UCL_S^{(\text{location})} = Q_{1-\alpha}\left(S_1^{(\text{location})}, \dots, S_B^{(\text{location})}\right)$.

Comparison of control limits

The complete-event simulation and location-stratified bootstrap limits were compared using their absolute differences: $\Delta_L = UCL_L^{(\text{location})} - UCL_L^{(\text{event})}$ and $\Delta_S = UCL_S^{(\text{location})} - UCL_S^{(\text{event})}$. The corresponding percentage differences, using the complete-event simulation limits as the reference, were calculated as $\Delta_L(\%) = 100\frac{UCL_L^{(\text{location})} - UCL_L^{(\text{event})}}{UCL_L^{(\text{event})}}$ and $\Delta_S(\%) = 100\frac{UCL_S^{(\text{location})} - UCL_S^{(\text{event})}}{UCL_S^{(\text{event})}}$.

Because the sampling locations were generated independently under the specified model, the location-stratified bootstrap was expected to approximate the complete-event simulation. Any observed differences between the two procedures can be attributed primarily to the finite size of the 100-event historical reference dataset and Monte Carlo variation in estimating the extreme $(1 - 1/740)$ quantile.

A fixed random-number seed was used to ensure reproducibility.

The following output shows stratified location level bootstrap UCL is more conservative than event level bootstrap UCL

	Statistic	UCL_event	UCL_location	Delta	Percent_Delta
1	L	3.852605	3.686554	-0.16605081	-4.310091
2	S	2.912009	2.859061	-0.05294814	-1.818268

Multivaraite Simulation

Simulation Algorithm for Comparing Event-Level and Location-Stratified Bootstrapping

A simulation study was conducted to compare control limits obtained from event-level statistics and location-stratified bootstrapping under a correlated-location model.

Step 1. Generate historical events

Generate 100 complete historical events from the random-effects model $X_{ir} = \mu_r + E_i + \varepsilon_{ir}$, where $i = 1, \dots, 100$ denotes the event index and $r = 1, \dots, 10$ denotes the sampling location.

The location means were specified as $\mu_r \in \{1.1, 1.2, \dots, 2.0\}$. The event effects and residual errors were generated independently from $E_i \sim N(0, 1)$ and $\varepsilon_{ir} \sim N(0, 3)$. Consequently, each location had marginal variance $\text{Var}(X_{ir}) = 4$ and standard deviation 2, while the within-event correlation between any two locations was $\text{Corr}(X_{ir}, X_{is}) = 0.25$.

Step 2. Calculate the observed event statistics

For each of the 100 simulated events, calculate $L_t = \text{median}(X_{t1}, \dots, X_{t10})$ and $S_t = \text{MAD}(X_{t1}, \dots, X_{t10})$, where MAD denotes the unscaled median absolute deviation. This produced 100 observed values of L_t and 100 observed values of S_t.

Step 3. Construct empirical location strata

Using the 100 complete historical events, construct ten empirical location strata, $G_1, \dots, G_{10}$, where G_r contains all observations collected from location r. Each stratum therefore contained 100 observations.

Step 4. Perform location-stratified bootstrapping

For each bootstrap replication:

1. Sample one observation with replacement from each stratum G_r.
2. Combine the ten sampled observations into a synthetic event.
3. Calculate the corresponding bootstrap statistics $L_t^{(b)}$ and $S_t^{(b)}$.

Repeat this procedure 200,000 times.

The resulting bootstrap distributions were used to estimate $UCL_{L,\text{Location}}$ and $UCL_{S,\text{Location}}$ at the probability level $1 - 1/740$.

Step 5. Perform event-level bootstrapping

For each bootstrap replication:

1. Sample one of the 100 observed events with replacement.
2. Record the corresponding observed event statistics L_t and S_t.

Repeat this procedure 200,000 times. The resulting bootstrap distributions were used to estimate $UCL_{L,\text{Event}}$ and $UCL_{S,\text{Event}}$ at the probability level $1 - 1/740$.

Step 6. Compare control limits

The two approaches were compared using $\Delta_L = UCL_{L,\text{Location}} - UCL_{L,\text{Event}}$ and $\Delta_S = UCL_{S,\text{Location}} - UCL_{S,\text{Event}}$. Percentage differences were calculated as $\Delta_L\% = 100 \times \Delta_L / UCL_{L,\text{Event}}$ and $\Delta_S\% = 100 \times \Delta_S / UCL_{S,\text{Event}}$.

Negative values indicate that the location-stratified bootstrap produced a more conservative control limit than the event-level bootstrap.

This gives $X_{ir} = \mu_r + E_i + \varepsilon_{ir}$, where $E_i \sim N(0,1)$ and $\varepsilon_{ir} \sim N(0,3)$. Therefore, $\text{Var}(X_{ir}) = 1+3 = 4$ and the marginal standard deviation remains $\text{SD}(X_{ir}) = 2$.

For two different locations $r \neq s$ from the same event, $\text{Cov}(X_{ir}, X_{is}) = 1$ and $\text{Corr}(X_{ir}, X_{is}) = \frac{1}{4} = 0.25$.

Thus, the simulation preserves the original marginal variance 2^2 while introducing moderate positive within-event correlation.

Table 2: Event-level and location-stratified bootstrap control limits under a correlated-location normal model.

Stat.	UCL_{event}	UCL_{loc}	Δ	$\Delta\%$
L	4.5939	3.7933	-0.8006	-17.4282
S	2.2731	2.8422	0.5691	25.0372

Important statistical interpretation

There is one important limitation that should be stated correctly.

With only 100 observed event-level values, the event bootstrap distribution contains only the original 100 distinct L_t values and 100 distinct S_t values. Resampling 200,000 times does **not** create new values. Because $1 - \frac{1}{740} = 0.9986486$,

the estimated event-level UCL will generally be the observed maximum: $UCL_{L,\text{event}} \approx \max_{t=1,\ldots,100} L_t$, and $UCL_{S,\text{event}} \approx \max_{t=1,\ldots,100} S_t$.

By contrast, location-stratified resampling can create many new combinations of location values. Therefore, it produces a much richer synthetic-event distribution. This difference is not a coding problem. It is an inherent consequence of estimating a 0.99865 quantile from only 100 complete events using a nonsmoothed empirical event-level bootstrap.